



/Users/halpern/teaching/352/labs/manual/master/rev_2022/skin_prelab.tex

Skin Depth Prelab

The two relevant Maxwell equations for an H field near or in a conductor are

$$\begin{aligned}\vec{\nabla} \times \vec{H} &= \vec{J} + \epsilon \frac{\partial \vec{E}}{\partial t} \\ \vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} = -\mu \frac{\partial \vec{H}}{\partial t},\end{aligned}\tag{1}$$

where the current density arises from E and non-zero conductivity, $\vec{J} = \sigma \vec{E}$.

For an initial field $H = \hat{z}H_z$:

First, convince yourself of two limiting cases:

If nothing varies in time, $\partial H_z / \partial t = 0$.

If $\sigma = 0$ there is no current and these equations lead to perpendicular E and H travelling as a wave.

Now, assuming that E and H are harmonic, $\vec{H} = H_z e^{i(k_x x - \omega t)}$, find the dispersion relation, the relation between k^2 and ω^2 implied by Maxwell's equations. You should find that k^2 is complex when $\sigma \neq 0$.

Find the real and imaginary parts of k (not of k^2 !!). Which of these corresponds to a penetration depth for H into a conductor?

